Basic Clustering Methods

- We use **clustering in unsupervised machine learning** to group **unlabeled data points** into natural structures or similarities without needing pre-existing answers or training labels

Key reasons for using clustering:

- Discovering Hidden Patterns
- Working with Unlabeled Data
- Customer & Market Segmentation
- Anomaly & Fraud Detection
- Data Preprocessing & Compression
- Document & Image Organization

Clustering

Classification vs clustering

S.No.	Clustering	Classification
1.	Unsupervised learning and cluster formation are done by trial and error as there is no supervisor	Supervised learning with the presence of a supervisor to provide training and testing data
2.	Unlabelled data	Labelled data
3.	No prior knowledge in clustering	Knowledge of the domain is must to label the samples of the dataset
4.	Cluster results are dynamic	Once a label is assigned, it does not change

Clustering advantages and disadvantages

Adv. and disadv. of clustering schemes

S.No.	Advantages	Disadvantages
1.	Cluster analysis algorithms can handle missing data and outliers.	Cluster analysis algorithms are sensitive to initialization and order of the input data.
2.	Can help classifiers in labelling the unlabelled data. Semi-supervised algorithms use cluster analysis algorithms to label the unlabelled data and then use classifiers to classify them.	Often, the number of clusters present in the data have to be specified by the user.
3.	It is easy to explain the cluster analysis algorithms and to implement them.	Scaling is a problem.
4.	Clustering is the oldest technique in statistics and it is easy to explain. It is also relatively easy to implement.	Designing a proximity measure for the given data is an issue.

Applications of clustering schemes

1. Grouping based on customer buying patterns
2. Profiling of customers based on lifestyle
3. In information retrieval applications (like retrieval of a document from a collection of documents)
4. Identifying the groups of genes that influence a disease
5. Identification of organs that are similar in physiology functions
6. Taxonomy of animals, plants in Biology
7. Clustering based on purchasing behaviour and demography
8. Document indexing
9. Data compression by grouping similar objects and finding duplicate objects

Clustering advantages and disadvantages

Challenges of clustering schemes

- Huge collection of data with higher dimensions i.e features or attributes can be a problem for clustering algorithm
- Scaling issues for such huge data
- Scaling is also an a issue where algorithms work with lower dimension data , but do not perform well for higher dimension data
- Units of data: like weight in kgs. and sometimes in pounds can pose problem for clustering

What is proximity measure?

- Clustering needs to find a measure to find similarity and dissimilarity among objects to group them
- Similarity and dissimilarity are collectively known as proximity measures
- Usually distance measures are used to find the similarity between 2 objects say i and j
- Dissimilarity measures: indicates how one object is different from another
- Distance between 2 objects say I and j, is denoted by D_{ij}

Property of distance measure is:

1. D_{ij} is always positive or zero.
2. $D_{ij} = 0$, i.e., the distance between the object to itself is 0.
3. $D_{ij} = D_{ji}$. This property is called symmetry.
4. $D_{ij} \leq D_{ik} + D_{kj}$. This property is called triangular inequality.

If all these conditions are satisfied, then the distance measure is called a metric.

Some proximity measures

- Euclidean distance
- City block distance
- Chebyshev distance
- Minkowski distance
- Binary attributes
- Simple matching coefficient(SMC)
- Jacquard coefficient
- Hamming distance
- Vector type distance measures

Euclidean distance

- One of the most important common distance measures
- Also called L_2 norm
- Defined as square root of squared differences between the coordinates of a pair of objects
- Euclidean distance between objects x_i and x_j with k features is :

$$\text{Distance}(x_i, x_j) = \sqrt{\sum_{k=1}^{n}(x_{ik} - x_{jk})^2} \qquad (13.1)$$

Minkowski distance

- Can be calculated as:

$$\text{Distance}(x_i, x_j) = \left(\sum \left| x_{ik} - x_{jk} \right|^r \right)^{\frac{1}{r}}$$

- If r=1: distance measure is called city block distance
- If r=2, distance measure is called Euclidean distance
- If r=infinity, distance measure is called Chebyshev distance

HIERARCHICAL CLUSTERING ALGORITHMS

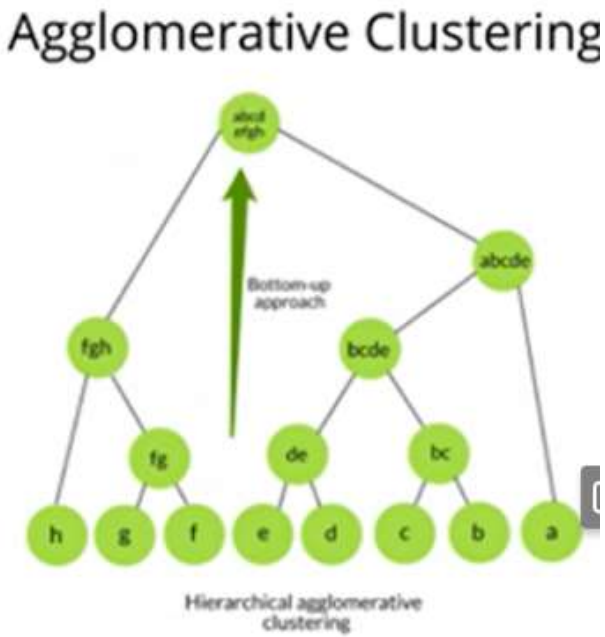
- Hierarchical methods produce nested partition of objects with Hierarchical relationships among objects
- Hierarchical relationships is shown in form of a dendrogram
- Hierarchical methods include:
  - Categories
  - Agglomerative methods: initially all individual samples are considered as cluster ie. A cluster with single element. Then they are merged to form single cluster
  - Divisive methods: a single cluster of all samples of dataset is taken initially and then partitioned. Partition is continued till smaller clusters are formed

Procedure for agglomerative algorithms

Algorithm 13.1: Agglomerative Clustering

1. Place each N sample or data instance into a separate cluster. So, initially N clusters are available.
2. Repeat the following steps until a single cluster is formed:
   (a) Determine two most similar clusters.
   (b) Merge the two clusters into a single cluster reducing the number of clusters as N–1.
3. Choose resultant clusters of step 2 as result.

- It follows bottom-up approach
- treats each data as a singleton cluster at the outset
- successively agglomerate pairs of clusters until all clusters have been merged into a single cluster that contains all data



TYPES OF LINKAGE IN HIERARCHICAL CLUSTERING

- **Hierarchical clustering** is used to group similar data points and organize data in a tree-like structure.
- Key part of this process is **linkage** which **calculates the distance between clusters before they are merged or divided**.
- Different types of linkage is used measure this distance differently
- TYPES:
  - **Single Linkage**
  - **Complete Linkage**
  - **Average Linkage**

SINGLE VS COMPLETE VS AVERAGE LINKAGE

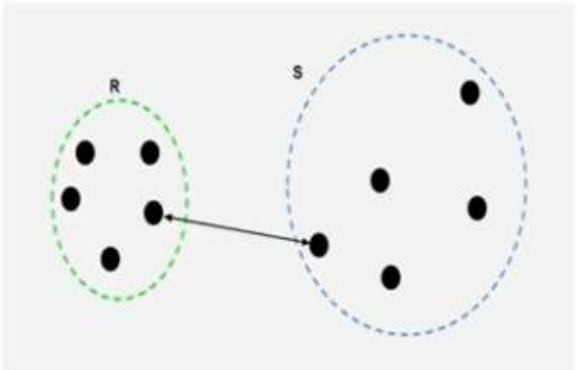
1. Single Linkage

For two clusters **R** and **S** the **single linkage returns the minimum distance between two points.**
This method creates long, chain-like clusters because it is **sensitive to outliers** and can **connect clusters based on a very small number of close points.**

$$L(R,S) = min(D(i,j)), i \epsilon R, j \epsilon S$$

where

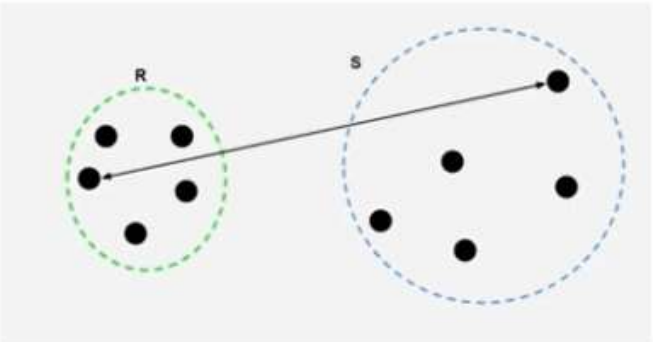
- **D(i, j)**: Distance function between points **i** and **j**.



2. **Complete Linkage**

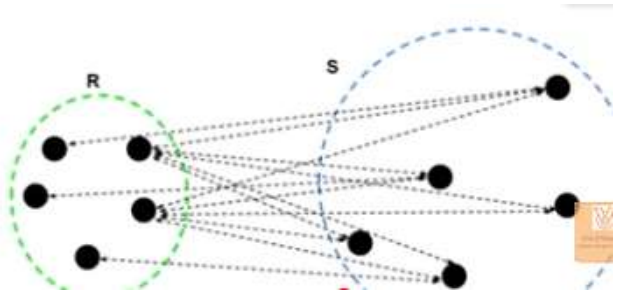
For two clusters **R** and **S** the **complete linkage returns the maximum distance between two points.**
It tends to create compact and spherical clusters because it is **more sensitive to outliers and tries to make sure that the clusters are not too far.**

$$L(R,S) = \max(D(i,j)),\ i \in R,\ j \in S$$



3. Average Linkage

It returns the **average distance between all pairs of points from two clusters.**
This method maintain a **balance between single and complete linkage** by considering all pairs of points not just the closest or farthest point.
It usually **results in clusters that are moderately compact.**



Single linkage or MIN algorithm

- Hierarchical clustering: produces nested clusters which can be visualized as hierarchical tree or dendrogram
- Idea: proximity among clusters
- In single linkage, the smallest distance (x,y), where x is from one cluster and y is from another cluster
- Distance between all possible pairs of 2 groups or clusters
- Used for merging the clusters
- This corresponds to finding a minimum spanning tree of a graph

Problem on Hierarchical Clustering (using Single Linkage)

Consider the array of points as shown in the following Table. 13.4.

Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

- **Step 1: Compute proximity matrix using Euclidean distance as shown in Table 13.5**

$D(0,1)=\sqrt{(1-2)^2+(4-8)^2} = 4.123$

$D(0,2)=\sqrt{(1-5)^2+(4-10)^2} = 7.211$

$D(0,3)=\sqrt{(1-12)^2+(4-18)^2} = 17.804$

$D(0,4)=\sqrt{(1-14)^2+(4-28)^2} = 27.295$

$D(1,2)=\sqrt{(2-5)^2+(8-10)^2} = 3.606$

$D(1,3)=\sqrt{(2-12)^2+(8-18)^2} = 14.142$

$D(1,4)=\sqrt{(2-14)^2+(8-28)^2} = 23.324$

$D(2,3)=\sqrt{(5-12)^2+(10-18)^2} = 10.630$

$D(2,4)=\sqrt{(5-14)^2+(10-28)^2} = 20.124$

$D(3,4)=\sqrt{(12-14)^2+(18-28)^2} = 10.198$

Given Table

Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Consider the array of points as shown in the following Table. 13.4.

Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Table 13.5: Proximity Matrix

Objects	0	1	2	3	4
0	–	4.123	7.211	17.804	27.295
1		–	3.606	14.142	23.324
2			–	10.630	20.124
3				–	10.198
4					–

- From the proximity table 13.5, we can observe that the minimum distance is 3.606. Therefore, we can group items 1 and 2 and cluster it together

Table 13.5: Proximity Matrix

Objects	0	1	2	3	4
0	–	4.123	7.211	17.804	27.295
1		–	3.606	14.142	23.324
2			–	10.630	20.124
3				–	10.198
4					–

Table 13.6: After Iteration 1

Clusters	{1, 2}	0	3	4
{1, 2}	–	4.123	10.630	20.124
0		–	17.804	27.295
3			–	10.198
4				–

- From the proximity table 13.5, we can observe that the minimum distance is 3.606. Therefore, we can group items 1 and 2 and cluster it together
- After iteration 1 (or after clustering 1&2 the proximity cluster table is as shown in figure 13.6

Table 13.5: Proximity Matrix

Objects	0	1	2	3	4
0	–	4.123	7.211	17.804	27.295
1		–	3.606	14.142	23.324
2			–	10.630	20.124
3				–	10.198
4					–

Table 13.6: After Iteration 1

Clusters	{1, 2}	0	3	4
{1, 2}	–	4.123	10.630	20.124
0		–	17.804	27.295
3			–	10.198
4				–

Thus, the distance between {1, 2} and {0} is:

$$\text{Minimum } \{\{1, 0\}, \{2, 0\}\} = \text{Minimum } \{4.123, 7.211\} = 4.123$$

The distance between {1, 2} and {3} is given as:

$$\text{Minimum } \{\{1, 3\}, \{2, 3\}\} = \text{Minimum } \{14.412, 10.630\} = 10.630$$

The distance between {1, 2} and {4} is given as:

$$\text{Minimum } \{\{1, 4\}, \{2, 4\}\} = \text{Minimum } \{23.324, 20.124\} = 20.124$$

Thus, the distance between {1, 2} and {0} is:

$$\text{Minimum } \{\{1, 0\}, \{2, 0\}\} = \text{Minimum } \{4.123, 7.211\} = 4.123$$

Table 13.6: After Iteration 1

Clusters	{1, 2}	0	3	4
{1, 2}	–	4.123	10.630	20.124
0		–	17.804	27.295
3			–	10.198
4				–

Now the minimum distance is: 4.123

Therefore, {0, 1, 2} is clustered together. This result is shown in Table 13.7.

Table 13.7: After Iteration 2

Clusters	{0, 1, 2}	3	4
{0, 1, 2}	–	10.630	20.124
3		–	10.198
4			–

After clustering {1,2} with 0

Thus, the distance between {0, 1, 2} and {3} using Eq. (13.10) is given as

Minimum {{0, 3}, {1, 3}, {2, 3}} = Minimum {17.804, 14.142, 10.630} = 10.630.

Thus, the distance between {0, 1, 2} and {4} is:

Minimum {{0, 4}, {1, 4}, {2, 4}} = Minimum {27.295, 23.324, 20.124} = 20.124.

Therefore, {0, 1, 2} is clustered together. This result is shown in Table 13.7.

Table 13.7: After Iteration 2

Clusters	{0, 1, 2}	3	4
{0, 1, 2}	–	10.630	20.124
3		–	10.198
4			–

After clustering {1,2} with 0

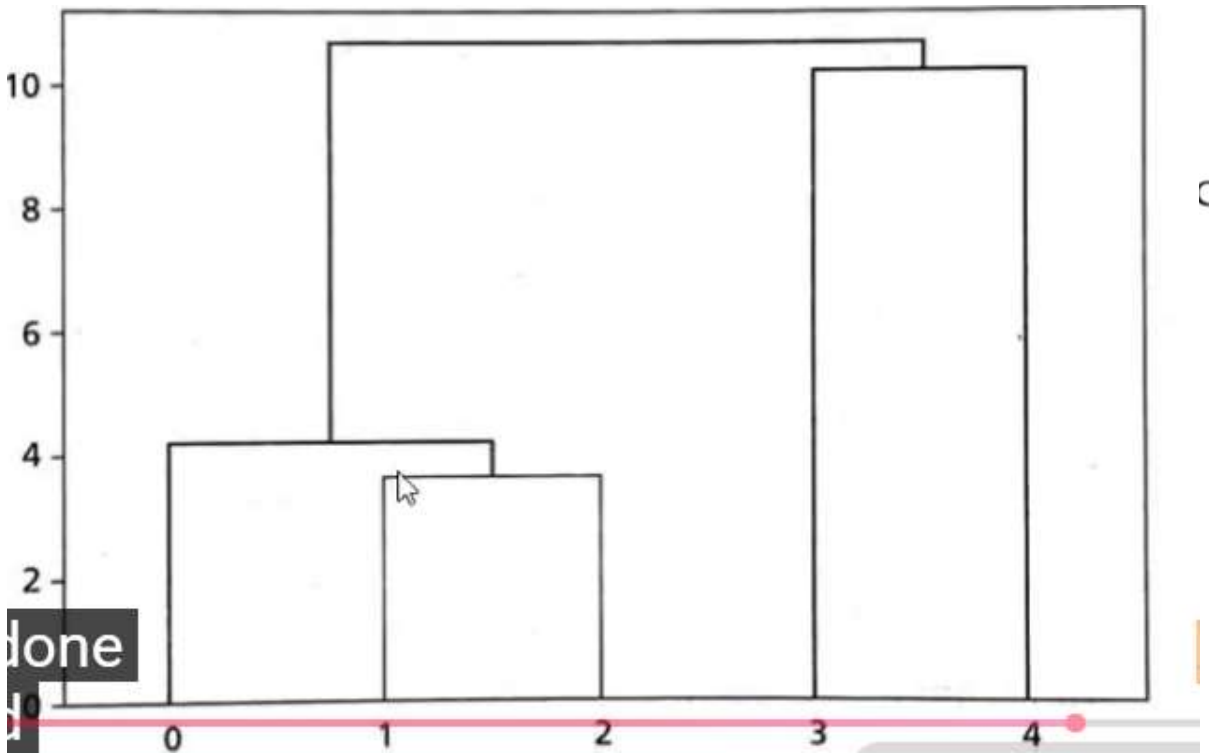
Therefore, the items {3, 4} are merged. And, the items {1, 2, 3} and {4, 5} are merged. The resultant is shown in Table 13.8.

Table 13.8: After Iteration 3

Clusters	{0, 1, 2}	{3, 4}
{0, 1, 2}		?
{3, 4}		–

The computation of ? in Table 13.8 is not needed as no other items are left. Therefore, the clusters {0, 1, 2} and {3, 4} are merged.

The final dendrogram for Table 13.4 is represented below :



Consider the array of points as shown in the following Table. 13.4.

Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

- The first minimum distance is 3.603 between 1&2
- Next cluster 0 with {1,2}at distance 4.123,
- Next cluster 3&4 at distance 10.198
- Finally {0,1,2} and {3,4} are merged

Basic Clustering Methods

- K-Means Clustering
- Expectation-Maximization Algorithm
- Gaussian Mixtures Clustering
- Hierarchical Clustering

Clustering Methods

1. Partitioning Methods: These methods **partition the objects into k clusters** and each partition forms one cluster.

- This method is used to optimize an objective criterion similarity function

  Ex: K-means

2. Hierarchical Based Methods: The **clusters formed in this method** form a **tree-type structure** based on the hierarchy.

- It is divided into two category :

A. **Agglomerative** (bottom-up approach)

B. **Divisive** (top-down approach)

- Ex: CURE (Clustering Using Representatives), BIRCH (Balanced Iterative Reducing Clustering and using Hierarchies), etc

Clustering Methods

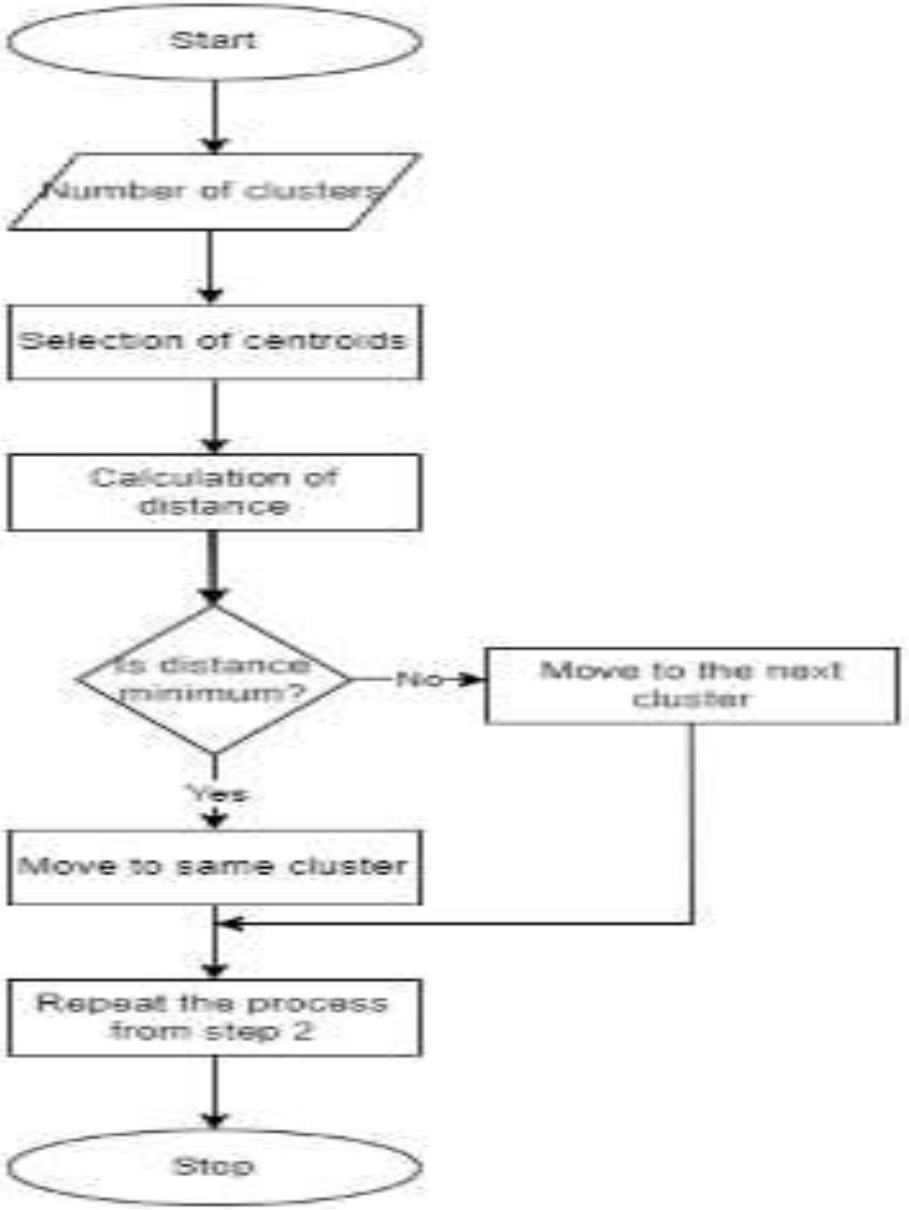
3. Density-Based Methods: These methods consider the clusters as the **dense region having some similarities** and differences from the lower dense region of the space.

Ex: DBSCAN (Density-Based Spatial Clustering of Applications with Noise), OPTICS (Ordering Points to Identify Clustering Structure), etc.

4. Grid-based Methods: In this method, the **data space is** formulated into a **finite number of cells** that form a **grid-like structure**.

Ex: STING (Statistical Information Grid), wave cluster, CLIQUE (CLustering In Quest), etc.

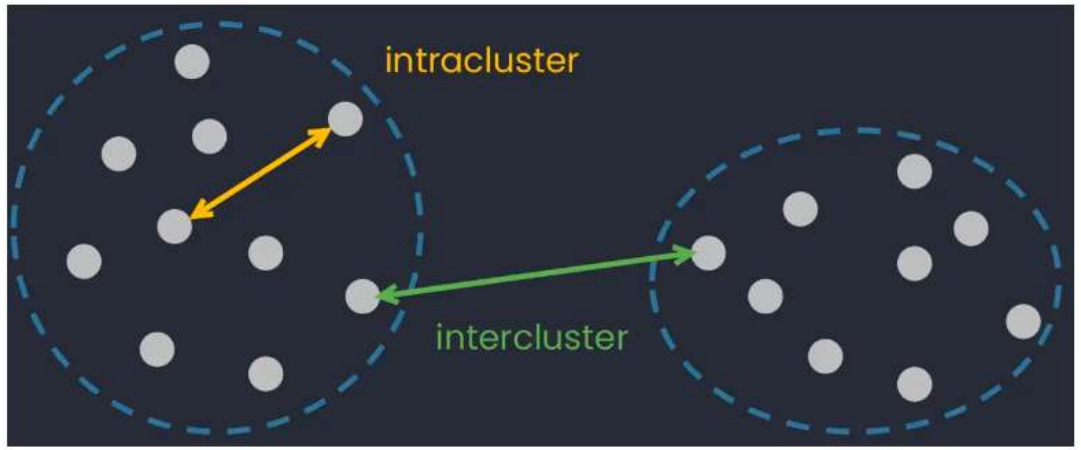
K-Means Clustering



intra-class similarity vs. inter-class similarity

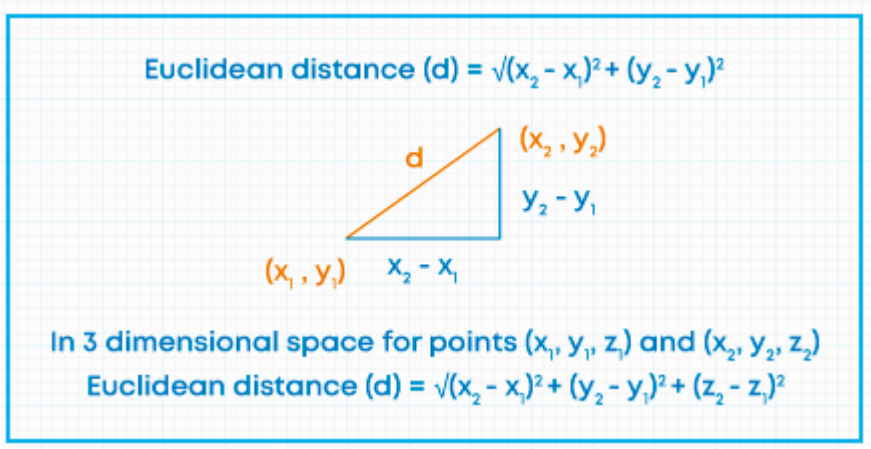
K-Means Algorithm Objective:

- Partition the data space in such a way that data points within the same cluster are as similar as possible (intra-class similarity),
- while data points from different clusters are as dissimilar as possible (inter-class similarity)
- Algorithm works through an iterative process until each data point is closer to its own cluster's centroid than to other clusters' centroids, minimizing intra-cluster distance at each step



Distance measure

- One way to determine similarity is through a measure of distance among the data points.
- The shorter the distance, the more similar the observations are.
- Here Euclidean distance is used.
- Consider two data observations over two attributes a and b. Point p1 has coordinates (a1,b1) and point p2=(a2,b2)
- The expression above can be extended to more than 2 attributes and the distance can be measured between any two points.



SSE – in detail

- The standard algorithm is the Hartigan-Wong algorithm (1979), which defines the total within-cluster variation as the sum of squared distances Euclidean distances between items and the corresponding centroid
- Within-Cluster-Sum of Squared Errors (WSS)
- Calculate the Within-Cluster-Sum of Squared Errors (WSS) for different values of k, and choose the k for which WSS becomes first starts to diminish. In the plot of WSS-versus-k, this is visible as an elbow

1. The Squared Error for each point is the square of the distance of the point from its predicted cluster center.
2. The WSS score is the sum of these Squared Errors for all the points.
3. Any distance metric like the Euclidean Distance or the Manhattan Distance can be used.

When to use KMeans?

- K means Clustering is used not to predict the labels but to actually find the labels in our data
- KMeans is best suited for:

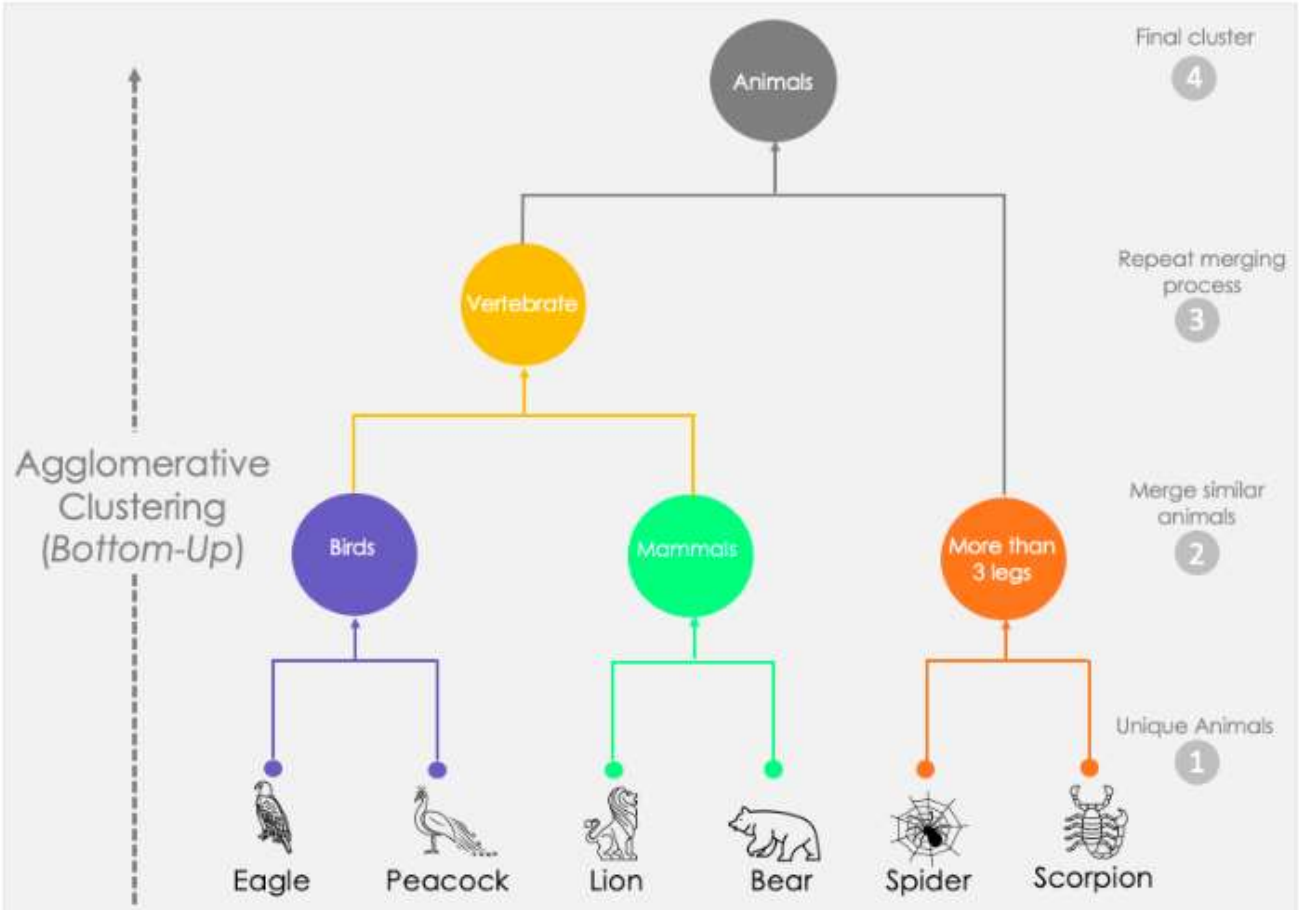
  - Unsupervised learning problems
  - With numeric data
  - does not work on categorical data
  - Better suited for linear datasets

Hierarchical clustering types

1. Agglomerative: Initially, each object is considered to be its own cluster. Clusters are then merged step by step until a single cluster remains. At the end of the cluster merging process, a cluster containing all the elements will be formed.

- Dendrogram – Visualization
- Agglomerative is popular method

Dendrogram of Agglomerative (bottom-up)



Hierarchical clustering types

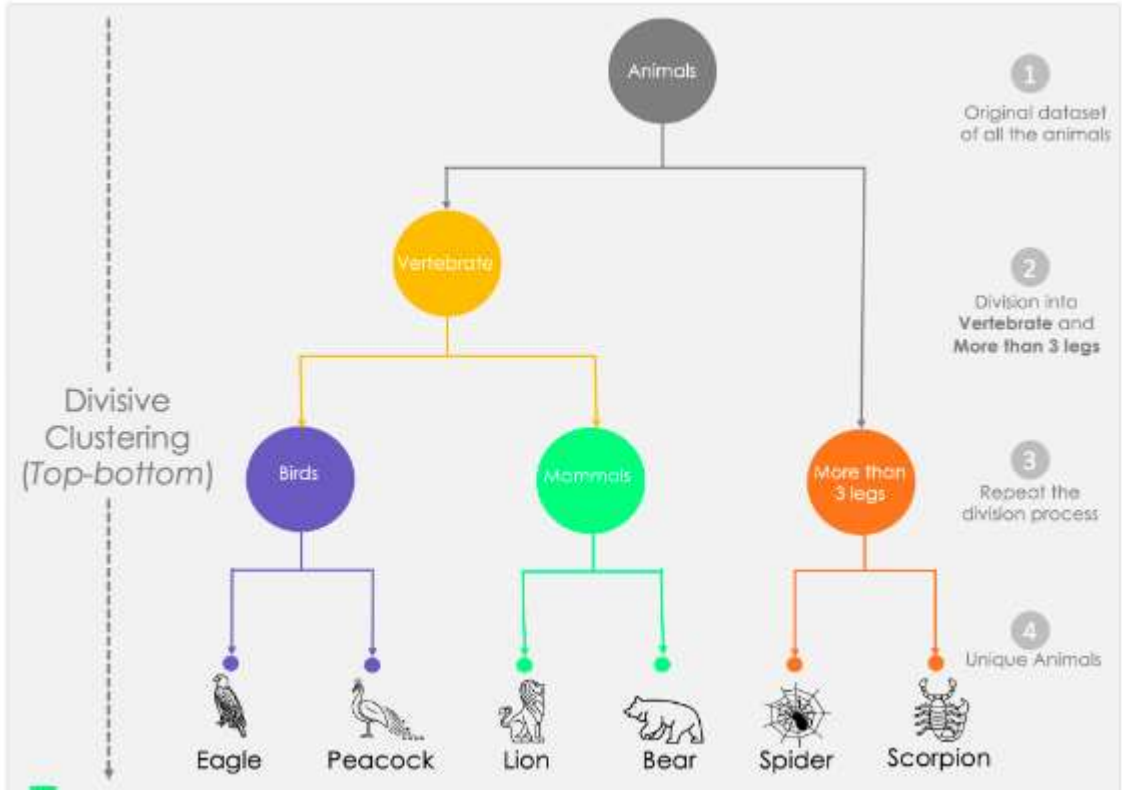
2. **Divisive**: The Divisive method is the **opposite of the Agglomerative method**.

Initially, *all objects are considered in a single cluster*.

Then the division process is performed step by step until each object forms a different cluster.

The cluster division or **splitting procedure is carried out according to some principles that maximum distance between neighboring objects** in the cluster.

Dendrogram of Divisive (top-down)



Agglomerative clustering

- Agglomerative clustering is a bottom-up approach.
- It starts clustering by treating the individual data points as a single cluster then it is merged continuously based on similarity until it forms one big cluster containing all objects.

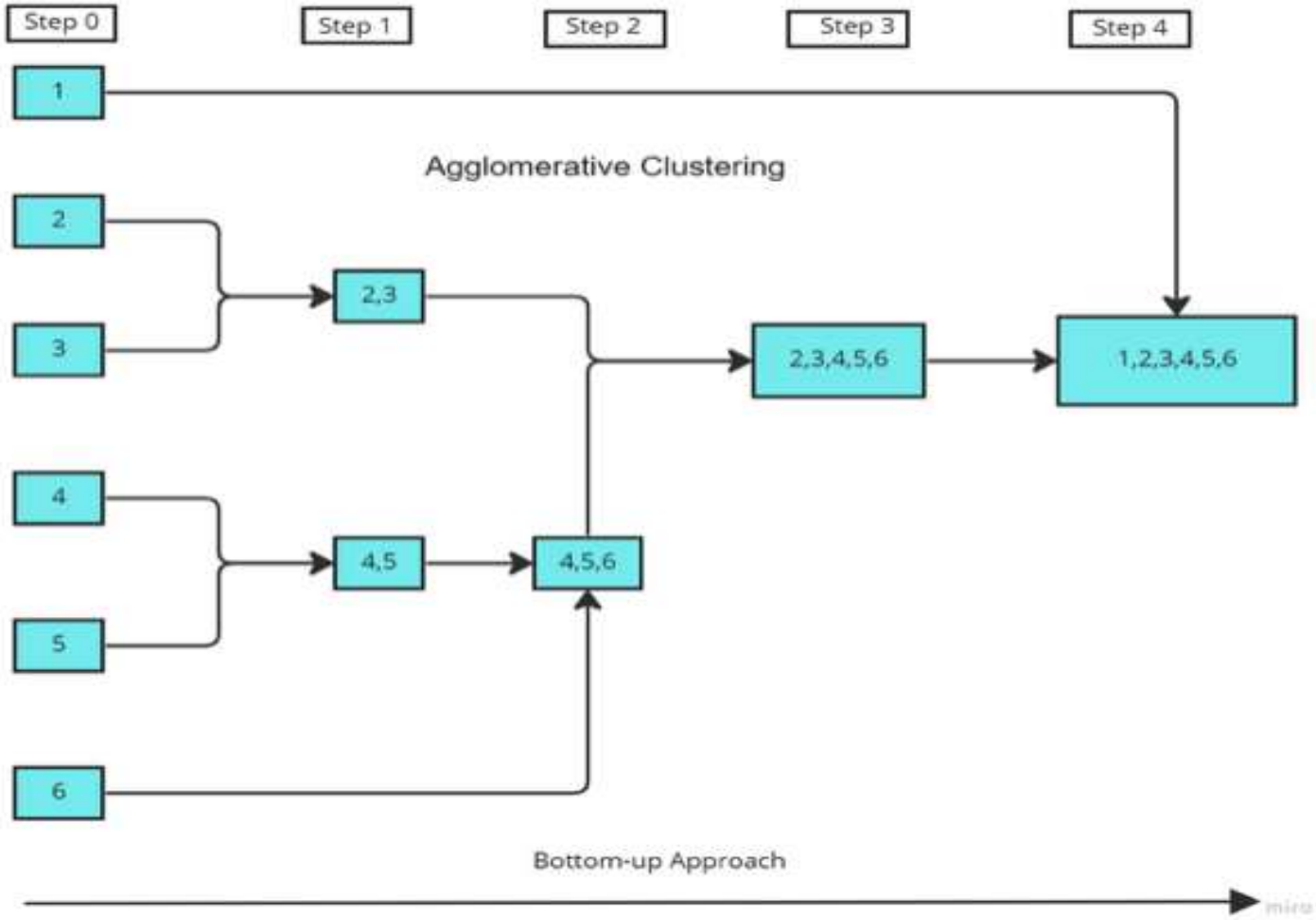
- The steps for agglomerative clustering are as follows:

1. **Compute the proximity matrix using a distance metric.**
2. **Use a linkage function to group objects into a hierarchical cluster tree based on the computed distance matrix from the above step.**
3. **Data points with close proximity are merged together to form a cluster.**
4. **Repeat steps 2 and 3 until a single cluster remains.**

Agglomerative clustering – Linkage Function

Linkage	*Description*
Complete	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *largest* of these dissimilarities.
Single	Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *smallest* of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time.
Average	Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *average* of these dissimilarities.
Centroid	Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B. Centroid linkage can result in undesirable *inversions*.

Agglomerative clustering - Pictorial Representation



Agglomerative clustering - PROXIMITY MATRIX

- The proximity matrix is a matrix consisting of the distance between each pair of data points.
- The distance is computed by a distance function.
- Euclidean distance is one of the most commonly used distance functions.
- proximity matrix consists of n points named x, and the $d(x_i,x_j)$ represents the distance between the points

	x_1	x_2	x_3	...	x_n
x_1	$d(x_1,x_1)$	$d(x_1,x_2)$	$d(x_1,x_3)$	...	$d(x_1,x_n)$
x_2	$d(x_2,x_1)$	$d(x_2,x_2)$	$d(x_2,x_3)$	...	$d(x_2,x_n)$
x_3	$d(x_3,x_1)$	$d(x_3,x_2)$	$d(x_3,x_3)$	...	$d(x_3,x_n)$
...	...	...	...	...	...
x_n	$d(x_n,x_1)$	$d(x_n,x_2)$	$d(x_n,x_3)$		$d(x_n,x_n)$

Agglomerative clustering – Linkage Function

- In order to group the data points in a cluster, a linkage function is used
- Values in the proximity matrix are taken and the data points are grouped based on similarity.

Agglomerative clustering- linkage methods

1. Single linkage: The **minimum** of all pairwise distance between elements in each pair of clusters is used to measure the distance between two clusters.
2. Complete linkage: The **maximum** of all pairwise distance between elements in each pair of clusters is used to measure the distance between two clusters.
3. Average linkage: The **average** of all pairwise distances between elements in each pair of clusters is used to measure the distance between two clusters.
4. Centroid linkage: Before merging, the distance between the two clusters' centroids are considered.
5. Ward's Method: It uses squared error to compute the similarity of the two clusters for merging.

Note: Different linkage methods lead to different clusters.

Single vs. Complete Linkage

Similarity of two clusters C1 and C2 = sim(c1, c2)
points Pi and Pj

$$\text{Sim}(C1,C2) = \text{Min Sim}(Pi,Pj) \text{ such that } Pi \in C1 \ \& \ Pj \in C2$$

$$\text{Sim}(C1,C2) = \text{Max Sim}(Pi,Pj) \text{ such that } Pi \in C1 \ \& \ Pj \in C2$$

Average vs. Wards Method

- Similarity of two clusters C1 and C2 = sim(c1, c2)
- points Pi and Pj
- Ward's Method: similarity between two clusters is exactly the same as Average linkage except that Ward's method calculates the sum of the square of the distances Pi and PJ
- Average vs Wards Method

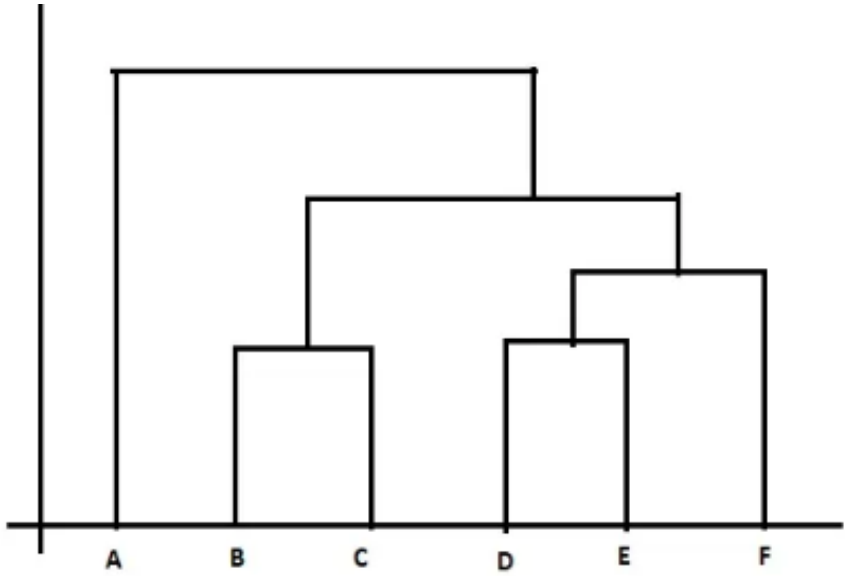
$$sim(C1,C2) = \sum sim(Pi, Pj)/|C1|*|C2|$$

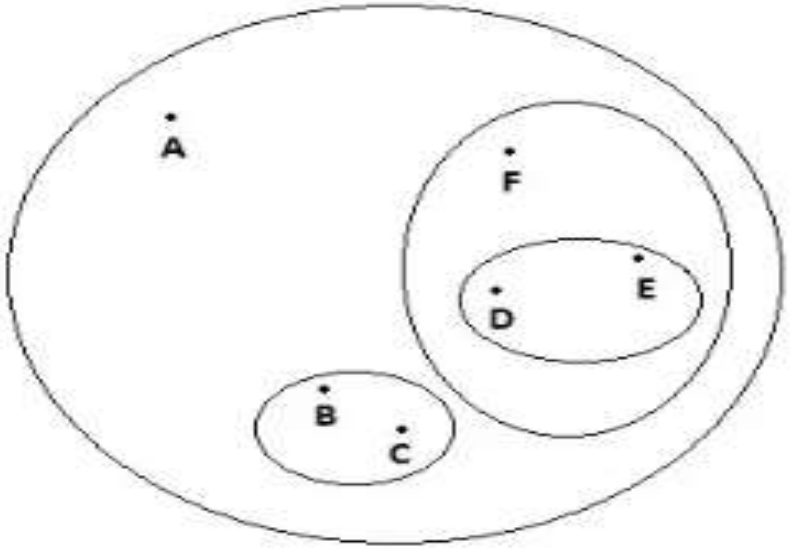
$$sim(C1,C2) = \sum (dist(Pi, Pj))^2/|C1|*|C2|$$

where, $Pi \in C1$ & $Pj \in C2$

Agglomerative clustering - Visualization

- The Hierarchical clustering Technique can be visualized using a Dendrogram.
- The below tree is a dendrogram of clusters shown





Hierarchical Clustering

Suppose we have the following data points in a two-dimensional space:

A: (2, 3)

B: (4, 5)

C: (6, 7)

D: (8, 9)

E: (10, 11)

To perform hierarchical clustering using the single-linkage method (also known as the nearest neighbor method).

We'll start by considering each point as an individual cluster and iteratively merge the closest clusters until we have a single cluster.

Hierarchical Clustering

Step 1: Calculate Pairwise Distances

Compute the Euclidean distance between each pair of data points:

- Distance between A and B:

$$\sqrt{(4-2)^2+(5-3)^2} = \sqrt{8} = 2.83$$

- Distance between A and C:

$$\sqrt{(6-2)^2+(7-3)^2} = \sqrt{32} = 5.66$$

- Distance between A and D:

$$\sqrt{(8-2)^2+(9-3)^2} = \sqrt{68} = 8.25$$

- Distance between A and E:

$$\sqrt{(10-2)^2+(11-3)^2} = \sqrt{128} = 11.31$$

Step 2: Merge Closest Clusters

We start by merging the two closest points: A and B (distance = 2.83).

- Merge AB: {(2, 3), (4, 5)}

Next, we calculate distances from the merged cluster AB to the remaining points:

- Distance between AB and C:

$\sqrt{(6-4)^2 + (7-5)^2} = 2.82$

Distance from AB and D

$\sqrt{(8-4)^2 + (9-5)^2} = 5.65$

Distance from AB and E

$\sqrt{(10-4)^2 + (11-5)^2} = 8.48$

Hierarchical Clustering

Step 3: Merge Again

The closest pair now is AB and C (distance = 3.61).

- Merge ABC: {(2, 3), (4, 5), (6, 7)}

Finally, we calculate the distance from ABC to D and E:

- Distance between ABC and D:

$$\sqrt{(8-6)^2 + (9-7)^2} = \sqrt{8} = 2.83$$

- Distance between ABC and E:

$$\sqrt{(10-6)^2 + (11-7)^2} = \sqrt{20} = 4.47$$

Step 4: Final Merge

The closest pair now is ABC and D (distance = 2.83).

- Merge ABCD: {(2, 3), (4, 5), (6, 7), (8, 9)}

Our final hierarchical clustering result consists of a single cluster containing all data points.

Summary

1. Initial clusters: A, B, C, D, E
2. Merge AB: {(2, 3), (4, 5)}
3. Merge ABC: {(2, 3), (4, 5), (6, 7)}
4. Merge ABCD: {(2, 3), (4, 5), (6, 7), (8, 9)}

Hierarchical Clustering with Complete Linkage

1.Initialization:

1. Start by considering each point as an individual cluster: {A}, {B}, {C}, {D}, {E}.

2.Pairwise Distances:

1. Calculate the Euclidean distance between each pair of clusters (using the complete linkage method, which considers the maximum distance between any two points in the clusters).

Table

Cluster Pair	Distance
AB	2.83
AC	5.66
AD	8.25
AE	11.31
BC	3.61
BD	5.39
BE	7.81
CD	2.83
CE	4.47
DE	2.83

3. Merge Closest Clusters:

1. The closest pair is CD (distance = 2.83).
2. Merge CD: {(2, 3), (8, 9)}

4. Update Distances:

1. Recalculate distances from the merged cluster CD to the remaining clusters:

Table

Cluster Pair	Distance
ACD	5.66
BCD	5.39
ECD	4.47
BCE	7.81
CDE	2.83

5.Next Merge:

1. The closest pair now is CDE (distance = 2.83).
2. Merge CDE: {(2, 3), (6, 7), (8, 9)}

6.Final Merge:

1. All points are now in a single cluster: {(2, 3), (4, 5), (6, 7), (8, 9), (10, 11)}

Hierarchical Clustering with Average Linkage

1.Initialization:

1. Start by considering each point as an individual cluster: {A}, {B}, {C}, {D}, {E}.

2.Pairwise Distances:

1. Calculate the Euclidean distance between each pair of clusters.
2. For average linkage, we'll compute the average distance between all pairs of points in the clusters.

Table

Cluster Pair	Distance
AB	2.83
AC	5.66
AD	8.25
AE	11.31
BC	3.61
BD	5.39
BE	7.81
CD	2.83
CE	4.47
DE	2.83

3. Merge Closest Clusters:

1. The closest pair is CD (distance = 2.83).
2. Merge CD: {(2, 3), (8, 9)}

4. Update Distances:

1. Recalculate distances from the merged cluster CD to the remaining clusters.
2. For each pair, compute the average distance between all points in the clusters.

Table

Cluster Pair	Distance
ACD	4.83
BCD	6.64
ECD	7.81
BCE	7.81
CDE	4.83

5. Next Merge:

1. The closest pair now is ACD (distance = 4.83).
2. Merge ACD: {(2, 3), (6, 7), (8, 9)}

6. Final Merge:

1. All points are now in a single cluster: {(2, 3), (4, 5), (6, 7), (8, 9), (10, 11)}

Hierarchical Clustering with Centroid Linkage

1.Initialization:

1. Start by considering each point as an individual cluster: {A}, {B}, {C}, {D}, {E}.

2.Pairwise Distances:

1. Calculate the Euclidean distance between each pair of clusters.
2. For centroid linkage, we'll compute the distance between the centroids (mean points) of the clusters.

3.Compute Centroids:

1. Calculate the centroids for each cluster:

- Centroid of {A}:

$$\text{Centroid}_A = \left(\frac{2+4}{2}, \frac{3+5}{2}\right) = (3,4)$$

- Centroid of {B}:

$$\text{Centroid}_B = \left(\frac{4+6}{2}, \frac{5+7}{2}\right) = (5,6)$$

- Centroid of {C}:

$$\text{Centroid}_C = \left(\frac{6+8}{2}, \frac{7+9}{2}\right) = (7,8)$$

- Centroid of {D}:

$$\text{Centroid}_D = \left(\frac{8+10}{2}, \frac{9+11}{2}\right) = (9,10)$$

- Centroid of {E}:

$$\text{Centroid}_E = (10,11)$$

4. Pairwise Distances (Centroids):

•Calculate the distances between centroids

Table

Cluster Pair	Distance
AB	2.83
AC	4.24
AD	6.71
AE	8.49
BC	2.83
BD	4.24
BE	6.71
CD	2.83
CE	4.24
DE	2.83

5. Merge Closest Clusters:

- The closest pair is AB (distance = 2.83).
- Merge AB: {(2, 3), (4, 5)}

6. Update Centroids:

- Recalculate the centroid for the merged cluster AB:
  - Centroid of AB:

$$\text{Centroid}_{AB} = \left(\frac{2+4}{2}, \frac{3+5}{2}\right) = (3, 4)$$

7. Next Merge:

- The closest pair now is CD (distance = 2.83).
- Merge CD: {(6, 7), (8, 9)}

8. Update Centroids:

- Recalculate the centroid for the merged cluster CD:
  - Centroid of CD:

$$\text{Centroid}_{CD} = \left(\frac{6+8}{2}, \frac{7+9}{2}\right) = (7, 8)$$

9. Final Merge:

- Merge AB and CD: {(2, 3), (4, 5), (6, 7), (8, 9)}

To solve hierarchical clustering manually using **Ward's method**,
we follow these steps:

Ward's Method:

Ward's method aims to minimize the variance within each cluster. In other words, it chooses the pair of clusters to merge in a way that results in the smallest increase in total within-cluster variance (i.e., the sum of squared deviations from the cluster's mean).

Steps:

1. **Calculate initial distances between points** (in this case, Euclidean distance).
2. **Identify the pair of points with the smallest distance** and merge them.
3. **Recalculate distances** between the new cluster and the remaining points (using Ward's method).
4. Repeat the process until all points are merged into one cluster.

Data Points:

- A: (2, 3)
- B: (4, 5)
- C: (6, 7)
- D: (8, 9)
- E: (10, 11)

Step 1: Compute Euclidean Distances between Points

We use the formula for Euclidean distance between two points (x_1, y_1) and (x_2, y_2):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Points	Distance
A-B	$\sqrt{(4-2)^2 + (5-3)^2} = \sqrt{4+4} = 2.83$
A-C	$\sqrt{(6-2)^2 + (7-3)^2} = \sqrt{16+16} = 5.66$
A-D	$\sqrt{(8-2)^2 + (9-3)^2} = \sqrt{36+36} = 8.49$
A-E	$\sqrt{(10-2)^2 + (11-3)^2} = \sqrt{64+64} = 11.31$
B-C	$\sqrt{(6-4)^2 + (7-5)^2} = \sqrt{4+4} = 2.83$
B-D	$\sqrt{(8-4)^2 + (9-5)^2} = \sqrt{16+16} = 5.66$
B-E	$\sqrt{(10-4)^2 + (11-5)^2} = \sqrt{36+36} = 8.49$
C-D	$\sqrt{(8-6)^2 + (9-7)^2} = \sqrt{4+4} = 2.83$
C-E	$\sqrt{(10-6)^2 + (11-7)^2} = \sqrt{16+16} = 5.66$
D-E	$\sqrt{(10-8)^2 + (11-9)^2} = \sqrt{4+4} = 2.83$

Step 2: Merge the Closest Pair of Points

The smallest distance is 2.83, occurring between several pairs (A-B, B-C, C-D, and D-E).

Let's arbitrarily merge **A and B** into a new cluster, call it C_1={A,B}.

Step 3: Recalculate Distances using Ward's Method

To calculate the new distances between cluster C_1 and the remaining points, we first calculate the centroid of the new cluster:

$$\text{Centroid of } C_1 = \left(\frac{2+4}{2}, \frac{3+5}{2}\right) = (3, 4)$$

Now, calculate the distances between the centroid of C_1 and the remaining points:

- $d(C_1, C) = \sqrt{(6-3)^2 + (7-4)^2} = \sqrt{9+9} = 4.24$
- $d(C_1, D) = \sqrt{(8-3)^2 + (9-4)^2} = \sqrt{25+25} = 7.07$
- $d(C_1, E) = \sqrt{(10-3)^2 + (11-4)^2} = \sqrt{49+49} = 9.90$

Step 4: Merge the Closest Pair Again

The closest distance now is 4.24, between C_1 and C. Merge them into a new cluster $C_2 = \{A, B, C\}$.

Step 5: Recalculate Distances Again

The centroid of C_2 is the average of the points in C_2:

$$\text{Centroid of } C_2 = \left(\frac{2+4+6}{3}, \frac{3+5+7}{3} \right) = (4, 5)$$

Now, calculate the distances between the new cluster C_2 and the remaining points:

- $d(C_2, D) = \sqrt{(8-4)^2 + (9-5)^2} = \sqrt{16+16} = 5.66$
- $d(C_2, E) = \sqrt{(10-4)^2 + (11-5)^2} = \sqrt{36+36} = 8.49$

Step 6: Merge the Closest Pair Again

The closest distance is 5.66, between C_2 and D. Merge them into a new cluster $C_3 = \{A, B, C, D\}$.

Step 7: Final Merge

The centroid of C_3 is:

$$\text{Centroid of } C_3 = \left(\frac{2+4+6+8}{4}, \frac{3+5+7+9}{4} \right) = (5, 6)$$

Now calculate the distance between C_3 and E:

- $d(C_3, E) = \sqrt{(10-5)^2 + (11-6)^2} = \sqrt{25+25} = 7.07$

Finally, merge C_3 and E into one cluster.

Result:

The hierarchical clustering forms as follows:

1. Merge A and B
2. Merge C_1 with C
3. Merge C_2 with D
4. Merge C_3 with E

This results in the final hierarchical structure.

Use the **distance matrix** in Table1 to perform single link and complete link hierarchical clustering. Show your results by drawing a dendogram. The dendogram should clearly show the order in which the points are merged.

Table 1 Distance matrix

	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

Algorithm 8.3 Basic agglomerative hierarchical clustering algorithm.

1: Compute the proximity matrix, if necessary.
2: **repeat**
3: Merge the closest two clusters.
4: Update the proximity matrix to reflect the proximity between the new cluster and the original clusters.
5: **until** Only one cluster remains.

For the single link or MIN version of hierarchical clustering, the proximity of two clusters is defined as the minimum of the distance (maximum of the similarity) between any two points in the two different clusters.

Steps:

- Using graph terminology, start with all points as singleton clusters.
- Add links between points one at a time (shortest links first).
- These single links combine the points into clusters.

	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

Combine P1 and P2:

$$dist(\{P1, P2\}, \{P3\}) = min\big(dist(P1, P3), dist(P2, P3)\big)$$
$$= min(0.41, 0.64)$$
$$= 0.41$$

$$dist(\{P1, P2\}, \{P4\}) = min\big(dist(P1, P4), dist(P2, P5)\big)$$
$$= min(0.55, 0.98)$$
$$= 0.55$$

$$dist(\{P1, P2\}, \{P5\}) = min\big(dist(P1, P5), dist(P2, P5)\big)$$
$$= min(0.35, 0.98)$$
$$= 0.35$$

	P12	P3	P4	P5
P12	0.00	0.41	0.55	0.35
P3	0.41	0.00	0.44	0.85
P4	0.55	0.44	0.00	0.76
P5	0.35	0.85	0.76	0.00

Combine P12 and P5:

$$dist(\{P12, P5\}, \{P3\}) = min\big(dist(P12, P3), dist(P5, P3)\big)$$
$$= min(0.41, 0.85)$$
$$= 0.41$$

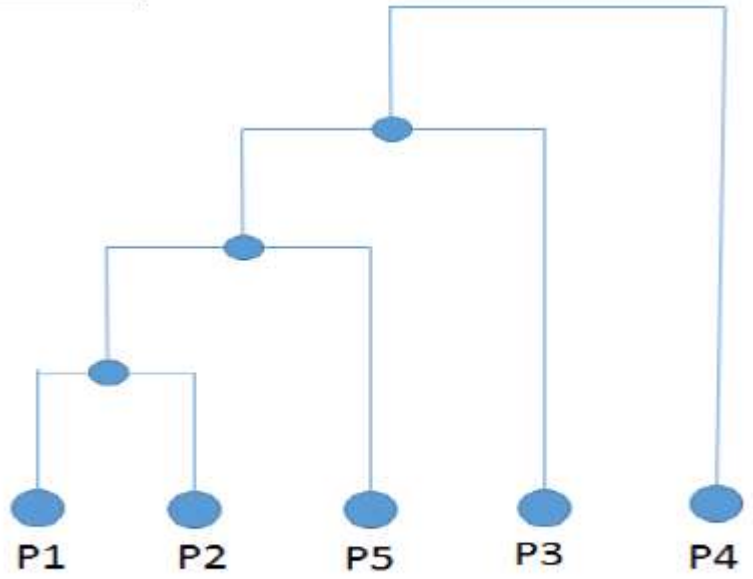
$$dist(\{P12, P5\}, \{P4\}) = min\big(dist(P12, P4), dist(P5, P4)\big)$$
$$= min(0.55, 0.76)$$
$$= 0.55$$

	P125	P3	P4
P125	0.00	0.41	0.55
P3	0.41	0.00	0.44
P4	0.55	0.44	0.00

Combine P125 and P3:

$$dist(\{P125, P3\}, \{P4\}) = min\big(dist(P125, P4), dist(P3, P4)\big)$$
$$= min(0.55, 0.44)$$
$$= 0.44$$

	P1235	P4
P1235	0.00	0.44
P4	0.44	0.00



Single Link Dendogram

For the complete link or MAX version of hierarchical clustering, the proximity of two clusters is defined as the maximum of the distance (minimum of the similarity) between any two points in the two different clusters.

Steps:

- Using graph terminology, start with all points as singleton clusters.
- Add links between points one at a time (shortest links first).
- Group points until all the points are completely linked, i.e., clique.

-	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

Combine P1 and P2:

$$dist(\{P1, P2\}, \{P3\}) = max\big(dist(P1, P3), dist(P2, P3)\big)$$
$$= max(0.41, 0.64)$$
$$= 0.64$$

$$dist(\{P1, P2\}, \{P4\}) = min\big(dist(P1, P4), dist(P2, P5)\big)$$
$$= min(0.55, 0.98)$$
$$= 0.98$$

$$dist(\{P1, P2\}, \{P5\}) = min\big(dist(P1, P5), dist(P2, P5)\big)$$
$$= min(0.35, 0.98)$$
$$= 0.98$$

	P12	P3	P4	P5
P12	0.00	0.64	0.98	0.98
P3	0.64	0.00	0.44	0.85
P4	0.98	0.44	0.00	0.76
P5	0.98	0.85	0.76	0.00

	P12	P3	P4	P5
P12	0.00	0.64	0.98	0.98
P3	0.64	0.00	0.44	0.85
P4	0.98	0.44	0.00	0.76
P5	0.98	0.85	0.76	0.00

Combine P3 and P4:

$$dist(\{P3, P4\}, \{P12\}) = min\big(dist(P3, P12), dist(P4, P12)\big)$$
$$= max(0.64, 0.98)$$
$$= 0.98$$

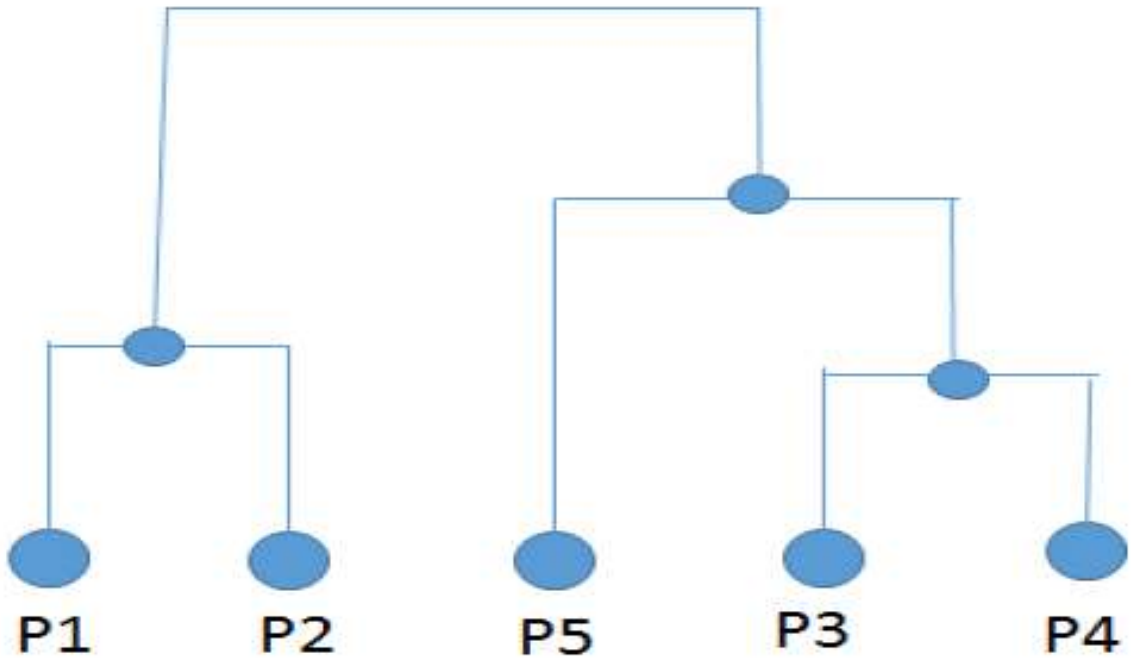
$$dist(\{P3, P4\}, \{P5\}) = min\big(dist(P3, P5), dist(P4, P5)\big)$$
$$= max(0.85, 0.76)$$
$$= 0.85$$

	P12	P34	P5
P12	0.00	0.98	0.98
P34	0.98	0.00	0.85
P5	0.98	0.85	0.00

Combine P34 and P5:

$$dist(\{P34, P5\}, \{P12\}) = max\big(dist(P34, P12), dist(P5, P12)\big)$$
$$= max(0.98, 0.98)$$
$$= 0.98$$

	P12	P345
P12	0.00	0.98
P345	0.98	0.00



Complete Link Dendogram

Use the **distance matrix** in Table2 to perform single link and complete link hierarchical clustering. Show your results by drawing a dendogram. The dendogram should clearly show the order in which the points are merged.

Table 2: Distance Matrix

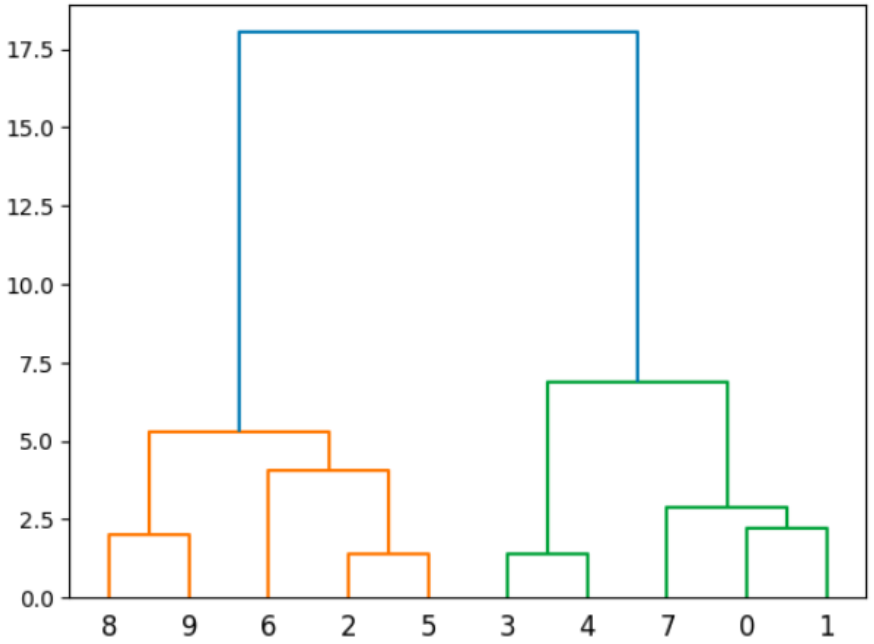
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Agglomerative clustering – implementation using Scipy

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram, linkage
x = [4, 5, 10, 4, 3, 11, 14 , 6, 10, 12]
y = [21, 19, 24, 17, 16, 25, 24, 22, 21, 21]
data = list(zip(x, y))
linkage_data = linkage(data, method='ward', metric='euclidean’)
dendrogram(linkage_data)
plt.show()
```

Note: default is ward linkage. Dendrogram graphing functions are not directly supported in sklearn. So used scipy here

Agglomerative clustering – implementation using Scipy



Agglomerative clustering - limitations

- High time and space complexity
  - n is the number of data points
  - Space complexity = $O(n^2)$ where n is the number of data points – store proximity matrix
  - Time complexity = $O(n^3)$ - to perform n iterations and in each iteration, to update the proximity matrix
- There is no objective function for hierarchical clustering
- Due to high time complexity, it cannot be used for large datasets.

Agglomerative clustering- Problem statement

- For the following 1-dimensional data points apply agglomerative hierarchical clustering to build the dendrogram. Construct the proximity matrix (distance matrix). Merge the clusters using the single linkage (min distance) and update the proximity matrix accordingly. Clearly show the proximity matrix corresponding to each iteration of the algorithm.

- 18, 22, 25, 27, 42, 43

Agglomerative clustering- 1D

Initially, every data point is considered a single-element cluster:
{18}, {22}, {25}, {27}, {42}, and {43}.

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

Agglomerative clustering- 1D

Iteration 1: Merge {42} and {43}

- **Minimum Distance**: The minimum non-zero value in the matrix is **1**, which is the distance between {42} and {43}.
- **Action**: Merge clusters {42} and {43} to form a new cluster **{42, 43}**.

	18	**22**	**25**	**27**	**{42, 43}**
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	**2**	17
27	9	5	**2**	0	15
{42, 43}	24	20	17	15	0

Iteration 2: Merge {25} and {27}

Minimum Distance: The minimum non-zero value in the current matrix is **2**, between {25} and {27}. [1]

	18	**22**	**{25, 27}**	**{42, 43}**
18	0	4	7	24
22	4	0	**3**	20
{25, 27}	7	**3**	0	15
{42, 43}	24	20	15	0

Iteration 3: Merge {22} and {25, 27}

Updated Proximity Matrix:

	18	{22, 25, 27}	{42, 43
18	0	4	24
{22, 25, 27}	4	0	15
{42, 43}	24	15	0

Iteration 4: Merge {18} and {22, 25, 27}

Updated Proximity Matrix:

	{18, 22, 25, 27}	{42, 43}
{18, 22, 25, 27}	0	15
{42, 43}	15	0

Agglomerative clustering- Problem statement

- For the following 2-dimensional data points apply agglomerative hierarchical clustering to build the dendrogram. Construct the proximity matrix (distance matrix). Merge the clusters using the single linkage (min distance) and update the proximity matrix accordingly. Clearly show the proximity matrix corresponding to each iteration of the algorithm.

Sample Number	X	Y
P1	0.4	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Agglomerative clustering- Problem statement

For the following 2-dimensional data points apply agglomerative hierarchical clustering to build the dendrogram. Construct the proximity matrix (distance matrix).

Merge the clusters using

a) complete linkage (max distance) and

b) average linkage

Update the proximity matrix accordingly. Clearly show the proximity matrix corresponding to each iteration of the algorithm.

Sample Number	X	Y
P1	1	1
P2	1.5	1.5
P3	5	5
P4	3	4
P5	4	4
P6	3	3.5

Expectation-Maximization Clustering

Expectation-Maximization Clustering

- We have many relevant features available to build our model
- But only a small portion of them are observable.
- We do not have the values for the not observed (latent) variables
- Expectation-Maximization algorithm tries to use the existing data to determine the optimum values for these variables and then finds the model parameters.

Expectation-Maximization Clustering

- Expectation-Maximization (EM) algorithm is a latent variable model
- A latent variable model consists of observable variables along with unobservable variables.
- Observed variables are those variables in the dataset that can be measured whereas
- unobserved (latent/hidden) variables are inferred from the observed variables.

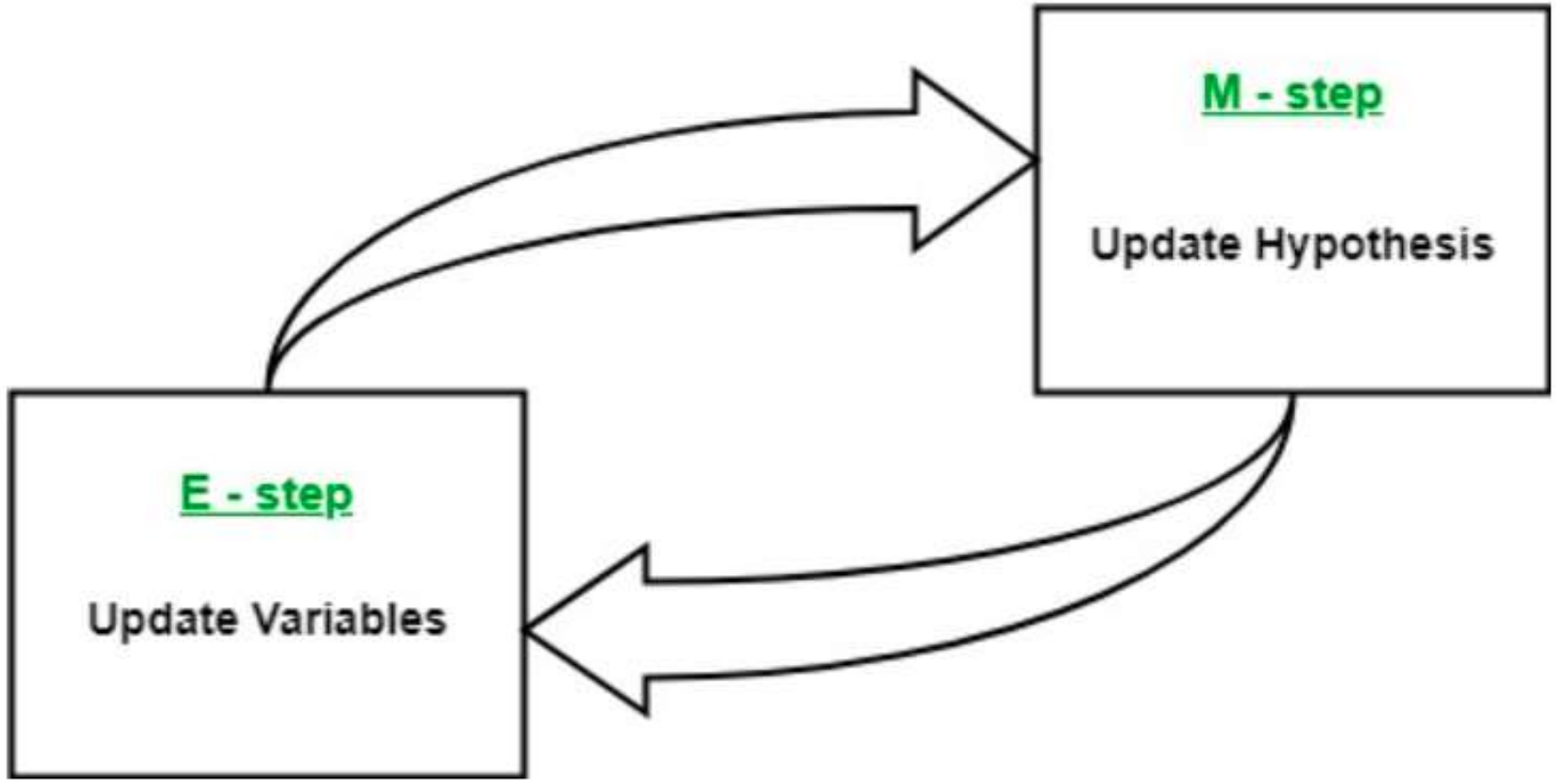
- It can be used to find the local maximum likelihood (MLE) parameters or maximum a posteriori (MAP) parameters for latent variables in a statistical or mathematical model.
- It is used to predict these missing values in the dataset, provided we know the general form of probability distribution associated with these latent variables.
- Basic idea behind this algorithm is to
  - use the observable samples of latent variables
  - to predict the values of samples that are unobservable for learning
  - This process is repeated until the convergence of the values occurs

Expectation-Maximization Clustering Algorithm

Algorithm:

1. Given a set of incomplete data, start with a set of initialized parameters.
2. Expectation step (E – step): In this expectation step, by using the observed available data of the dataset, try to estimate or guess the values of the missing data. Finally, after this step, we get complete data having no missing values.
3. Maximization step (M – step): Now, use the complete data, which is prepared in the expectation step, and update the parameters.
4. Repeat step 2 and step 3 until converge to solution

Expectation-Maximization Clustering

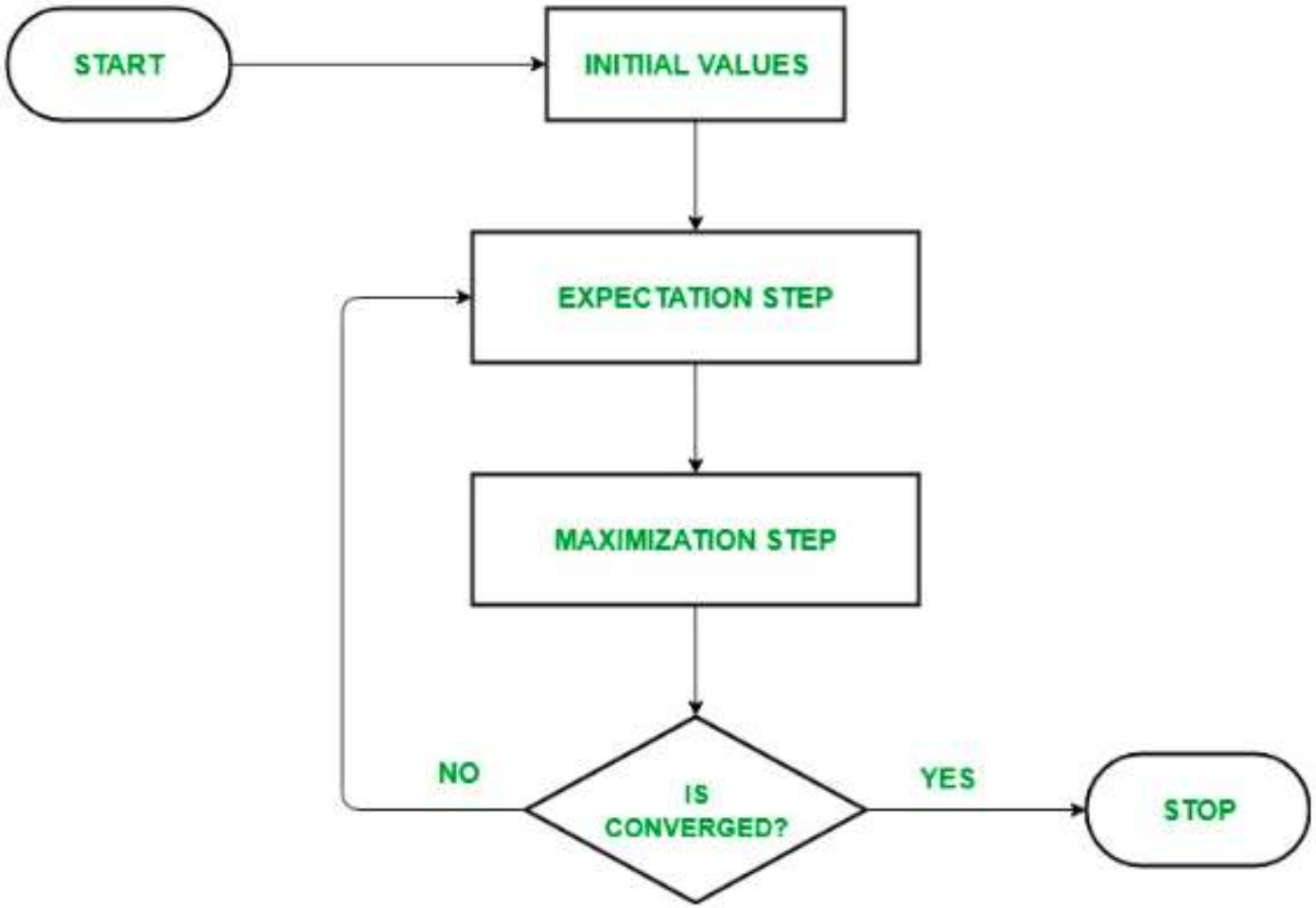


Expectation-Maximization Clustering Algorithm

Detailed Steps: The Expectation-Maximization algorithm aims to use the available observed data of the dataset to estimate the missing data of the latent variables and then using that data to update the values of the parameters in the maximization step.

1. Initialization Step: In this step, initialize the parameter values with a set of initial values, then give the set of incomplete observed data to the system with the assumption that the observed data comes from a specific model i.e, probability distribution.
2. Expectation Step: In this step, by using the observed data to estimate or guess the values of the missing or incomplete data. It is used to update the variables.
3. Maximization Step: In this step, use the complete data generated in the “Expectation” step to update the values of the parameters i.e, update the hypothesis.
4. Checking of convergence Step: Now, in this step, check whether the values are converging or not, if yes, then stop otherwise repeat these two steps i.e, the “Expectation” step and “Maximization” step until the convergence occurs.

Expectation-Maximization Clustering Algorithm

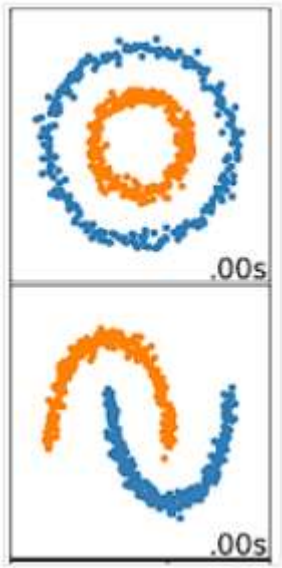


Expectation-Maximization Clustering

- K-Means calculates the mean of all the points in a cluster in every iteration, in order to proceed towards convergence.
- The Expectation Maximization Clustering algorithm is robust than K-Means, as it uses two parameters, Mean and Standard Deviation to define a particular cluster.

Disadvantages of K-Means

- Disadvantages of K-Means
- K-means clustering performs best on data that are spherical.
- Spherical data are data that group in space in close proximity to each other either. This can be visualized in 2 or 3 dimensional space more easily.
- Data that are not spherical or should not be spherical do not work well with k-means clustering.
- Ex: k-means clustering would not do well on the below data as we would not be able to find distinct centroids to cluster the two circles or arcs differently, despite them clearly visually being two distinct circles and arcs



K-Means vs Expectation-Maximization Clustering

"Hard" assignment vs "Soft" assignment

- K-Means - provides a cluster for each data point - doing "Hard" assignment
- Expectation-Maximization (EM) Clustering - Probabilistic generative model
- EM model provides the probability that a given data point belongs to each cluster.
- Instead of doing "hard" assignment, EM model performs "soft" assignment
- Each data point will belong to every cluster but with different probabilities

K-Means vs Expectation-Maximization Clustering

- Disadvantages of K-Means
- Lack of flexibility in cluster shape - tries to force-fit the data into circular clusters
- We can think k-means model places a circle (or, in higher dimensions, a hyper-sphere) at the center of each cluster, with a radius defined by the most distant point in the cluster.
- This radius acts as a hard cutoff for cluster assignment within the training set: any point outside this circle is not considered a member of the cluster
- k-means has no built-in way of accounting for oblong or elliptical clusters and for non-circular shapes and result in a mixing of cluster assignments where the resulting circles overlap
- Lack of probabilistic cluster assignment - for many datasets (especially low-dimensional datasets) it may not perform well
- EM algorithm is considered to be better than K-Means because it provides additional information about the data, namely, the dispersion (variance) of the cluster, not only its centers
- KMeans uses a distance-based approach, and GMM uses a probabilistic approach.

Expectation-Maximization Clustering

- Detecting similarities among thousands of records in a dataset is a burdensome task.
- Algorithms make different assumptions about the underlying probabilistic generative process of the data

Expectation-Maximization Clustering

- A single Gaussian or any other distribution with its mean and variance cannot map thousands of instances in a dataset into a set of K clusters accurately
- In the plot below, we would need at least 3 clusters to segment the data.
- In this context, Mixture Models (MM) are needed more specifically —Gaussian Mixture Models (GMM)

